

Q1. Blood is carried away from the heart by the arteries. The main artery is the aorta, which carries oxygenated blood to the whole body.

Q2. The chloroplast is the organelle mainly involved in photosynthesis. It contains chlorophyll which absorbs light energy.

Q5. Diagram of alveolus drawn below – alveolar sac is a small round air pocket, capillary wraps around it, oxygen moves in and carbon dioxide moves out.

Q3. Chloroplasts are the site of photosynthesis. Main pigments are chlorophyll a, chlorophyll b and carotenoids. The two stages are:

- 1. Light reaction – captures light energy and makes ATP and NADPH.
- 2. Dark reaction (Calvin cycle) – uses that energy to fix CO_2 into glucose.

Q6. The human digestive system starts at the mouth, then the food pipe (oesophagus) carries food to the stomach. The stomach churns food with acid and

(Q6 continued) enzymes. From the stomach food passes into the small intestine, where the liver sends bile and the pancreas sends pancreatic juice. Most absorption of digested food happens in the small intestine through villi. The remaining waste goes to the large intestine where water is absorbed.

Q7. Nephron diagram – Bowman's capsule surrounds the glomerulus, then the proximal tubule, then the loop of Henle goes down into the medulla, then the distal tubule, and finally the collecting duct carries urine away.

Q11 (a). Plant B is pale because in dim light the plant cannot make enough chlorophyll, so the leaves stay yellowish and stretch out to reach more light.

Q11 (b). Put Plant B near a sunny window.

Q14. Osmosis is the movement of water from a dilute solution to a concentrated solution through a semi-permeable membrane.

Q9. Transpiration is the loss of water as vapour from the leaves, mostly through the stomata. It pulls water up from the roots. Two factors that increase it are high temperature and wind (moving air).